LSHNet: Leveraging Structure-Prior With Hierarchical Features Updates for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— Salient object detection in optical remote sensing images (ORSI-SOD) is a task that detects the most prominent context in optical remote sensing images (ORSIs). ORSI-SOD is challenging due to the diverse sizes and shapes of the targets, their irregular distribution in the scene, and the occlusion caused by surrounding environments. Recently, many deep learning-based models have demonstrated promising performance in ORSI-SOD. However, there still remains considerable potential for addressing these challenges in ORSI-SOD. In this article, we propose a novel architecture, LSHNet. We propose a dual-branch architecture consisting of an edge encoder that leverages structure features using edges as the structure-prior and an image encoder that extracts context features from the image. We propose three modules. Image-structure fusion module (ISFM) integrates the two-stream features extracted from dual branch encoders through intrapatch and internal-patch attention mechanisms to utilize diverse receptive fields. Local–global feature fusion module (LGFM) transfers global features representing the target to local feature maps to discriminate the region of the targets from background clutters. The semantic cues updating module (SCUM) updates the representative features of the target from high-level to low-level. By integrating hierarchical information effectively, global features extracted from multilayers can be rectified. We experiment with the three main evaluated datasets in ORSI-SOD: ORSSD, EORSSD, and ORSI-4199. We demonstrate the promising results on the three datasets and analyze the effectiveness of the proposed modules in the ablation study.**

***Index Terms*— Dual branch architecture, feature refinement, optical remote sensing image (ORSI), patch attention, salient object detection.**

I. Introduction

SALIENT object detection in optical remote sensing image (ORSI-SOD) is a task within the field of computer vision that focuses on identifying and segmenting the most prominent contexts in optical remote sensing images (ORSIs). These contexts include water bodies, buildings, soil types, airplanes, roads, vegetation, and other significant land cover elements. Accurate detection of these features is crucial for various applications in environmental monitoring, urban planning, disaster management, and agriculture. ORSIs and natural scene images have significant differences. ORSIs are captured using remote sensors from a bird's-eye view such as satellites, aircraft, or drones to represent the Earth's surface including geographical, and infrastructural information. However, natural scene images are typically acquired from a frontal perspective using optical devices like cameras or smartphones and have information about environments visible to the human eye, such as everyday landscapes, urban settings, people, and animals. Due to the distinct characteristics of ORSIs compared to natural scene images, there has been a growing need for research focused on ORSIs. In addition to salient object detection, research focusing on optical remote sensing has also been conducted across various fields, including super-resolution [1], [2], [3], classification [4], [5], and learning strategies [6], [7]. In this article, we focus on addressing the challenges of salient object detection specifically in the context of ORSIs (ORSI-SOD).

Salient object detection is a field that has been extensively studied over a long period, initially focusing on natural scene images (NSI-SOD) in traditional methods [8], [9], [10]. With advancements in deep learning, significant progress has been made, resulting in impressive outcomes [11], [12], [13], [14]. Several notable studies proposed methods for capturing the saliency in natural scene images. GateNet [12] constructs gate units that control the dual information extracted from the encoder and decoder adaptively. EGNet [15] proposes a method for addressing the spatial coherence between two features by utilizing a module that predicts object-centric boundaries and another module that predicts semantic information. While numerous studies in NSI-SOD [11], [12], [14], [16], [17] have proposed effective methods for saliency extraction, many works in ORSI-SOD have validated that models of NSI-SOD show poor performance on ORSI datasets [18], [19], [20], [21], [22], [23]. As shown in Fig. 1, the characteristics of the target differ significantly between NSI-SOD and ORSI-SOD. The targets of ORSI-SOD have irregular

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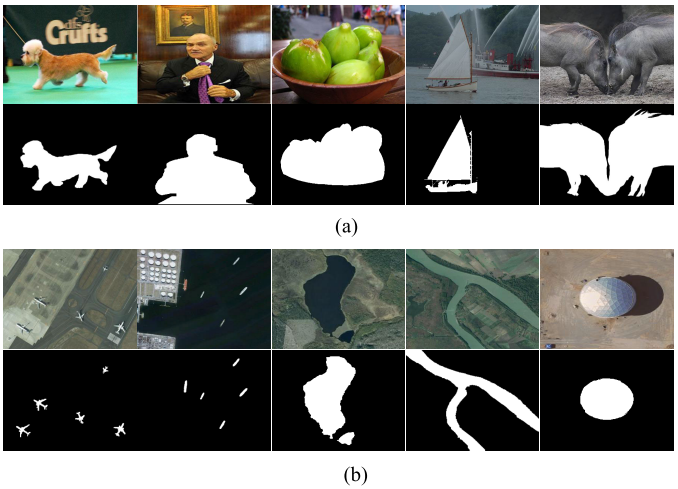


Fig. 1. Differences of NSI-SOD and ORSI-SOD. (a) Images and targets in NSI-SOD. (b) Images and targets in ORSI-SOD. The targets in the two domains differ significantly. The targets in ORSI-SOD vary in size and exhibit irregular distributions. In addition, their shapes, forms, and sizes are highly diverse depending on the type.

distributions, sizes, and shapes. Also, it suffers from complex background clutters. These challenges demonstrate that simply applying NSI-SOD methods to ORSI is not a trivial solution.

To address these challenging aspects, recent studies in ORSI-SOD have proposed various methods. MCCNet [20] introduces a module that simultaneously considers foreground, background, and edges. This module separately extracts features from the image and edges and then combines these features to enhance foreground–background discrimination, inspired by NSI-SOD [24]. ACCoNet [21] emphasizes utilizing adjacent contextual information to enhance saliency features across multiple layers. It introduces a module that interactively integrates high-level semantic features with low-level detailed features and employs a two-stream decoder structure to minimize interference. AESINet [22] uses separate modules for high-level and low-level features from multilayer CNNs. It captures adaptive edge information with perception maps from pooling differences and uses dilated convolutions for varied target sizes. High-level features are processed with graph convolution networks for deep semantic extraction and node attention.

As the observations mentioned above, we focus on leveraging edge information, refining features tailored to the semantics of the target, and integrating features across multiple layers. First, previous research [19], [20], [23] has only focused on using supervised learning to predict the edges of the annotated masks and using them as an additional loss function better to capture the edges of the ground truth (GT) masks. However, edges obtained from the image fundamentally provide crucial insights into the structure of the image, and they are valuable elements for understanding its overall context. Inspired by unsupervised video object segmentation (UVOS) works [25], [26], [27] and RGBD-SOD [28], [29], we propose a two-branch encoder architecture that leverages the edge obtained from the input image as structure-prior to extract deep structural features. Passing the RGB image and edge information through separate encoders, each feature can be extracted independently. It is effective to transfer structure attributes to image features. Second, previous studies [21], [22], [23] have relied on dilated convolution methods to increase receptive fields. However, this often results in overlooking local information due to the expanded receptive fields. We propose an image-structure fusion module (ISFM) to integrate the two-stream features extracted from dual branch encoders through intrapatch and interpatch attention mechanisms. We efficiently handle local fusion within patches and global fusion across patches through the intrapatch, internal-patch attention mechanism of ISFM. Third, previous works [20], [22], [30] refine feature maps through complex connectivity structures based on spatial attention (SA) and channel attention (CA) in multilayer features. We propose a local–global feature fusion module (LGFM) to transfer global features denoted as semantic cues (SCs) representing the target to local feature maps. We extract SCs representing the target and refine the feature maps centered around the target through these SCs. Fourth, we proposed a SCs updating module (SCUM) to update the representative features of the target from high-level to low-level across multilayers. Integrating hierarchical information sequentially, SCs extracted from multilayers can be rectified. In conclusion, we propose a network that leverages structural information with hierarchical information updates, LSHNet.

Our contributions are as follows.

1) We propose a new architecture that extracts information from both the image and edges in parallel using two encoders to utilize the deep structure fully features as structure-prior.
2) We designed a module that effectively integrates structural and context features by dividing them into patch internal and external units (ISFM).
3) We propose an LGFM that extracts the single attribute of the target features to filter out background clutters.
4) We propose a semantic features update module to fuse SCs of multilayer information hierarchically (SCUM).
5) Through comprehensive experiments on three major datasets: ORSSD, EORSSD, and ORSI4199, we demonstrate both quantitatively and qualitatively superior results. Also, we demonstrate the efficacy of our proposed method through ablation studies.

In the following sections, we discuss the abstract of studies that have been conducted over decades in Section II. Next, we elaborate on the methodological details of our proposed approach in Section III. Then, we compare ours with existing state-of-the-art works and demonstrate our proposed modules on ablation study in Section IV. Finally, we describe our contributions and their significance briefly in Section V.

II. Related Works

In this section, we discuss the various methods that have been researched in this field and the methodologies relevant to the proposed modules.

A. Salient Object Detection in Natural Scene Images

Before the advent of deep learning, research in this area encompassed various hand-crafted methodologies and techniques based on computational theory. Early studies

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demonstrate an unsupervised approach as images are segmented based on explicitly designed algorithms. Itti et al. [31] was a pioneer work that proposed the hand-crafted method to capture the saliency of the objects. Liu et al. [8] designed a tree structure for salient object detection. This work roughly segmented the image considering multiple image priors such as global contrast, object prior, and spatial sparsity. This segmentation map constructed a tree structure using the segmented regions as nodes. Yu et al. [32] proposed the probabilistic approach based on random walk theory and maximal entropy. Li et al. [33] and Yuan et al. [34] advanced by employing regularization and reverse correction techniques based on random walk. Kim et al. [35] introduced high-dimensional color transforms to expand the information of an image and local learning methods. However, these traditional methods had limitations in finding optimal solutions for various scenarios in ORSIs, leading to poor generalization performance.

CNN-based networks based on supervised learning in deep learning significantly surpassed traditional methods in salient object detection. Recently, many works [11], [12], [13], [14], [15], [36] have developed the performance of NSI-SOD. PoolNet [36] designed the base architecture in NSI-SOD which consists of a U-Net shape with FPN structure. This article expanded U-Net with pooling techniques. EGNet [15] proposed a network that utilizes edge guidance to capture explicit boundaries between foreground and background. By predicting the mask's edge output additionally, it considered not only the saliency region but also the boundaries of the target. GateNet [12] proposed the gate unit based on FPN architecture. This module adaptively determines the weighting of each feature extracted from the encoder and decoder. U2Net [13] designed a new architecture based on U-Net. The overall large U-Net structure was created by combining small encoder–decoder module U-Net structures into one block, resembling a fractal structure. This allows for the exploration of information across multiple resolutions even within a single small block. As mentioned in the introduction, due to the differences in characteristics between RSI-SOD and NSI-SOD datasets, these studies showed promising performance in NSI-SOD and provided excellent baselines for RSI-SOD, but showed inferior results in ORSI-SOD.

B. Salient Object Detection in ORSIs

Observations that NSI-SOD does not perform well in ORSI-SOD have prompted research efforts aimed at achieving better performance in the latter. CNN-based architectures derived from NSI-SOD methods have been proposed for ORSI-SOD. LVNet [18] introduced the ORSSD dataset and proposed a nested network with a two-stream pyramid architecture which has an L-V-shaped structure to be applied to ORSI-SOD. The ORSSD dataset serves as the most commonly used benchmark dataset for evaluating the performance of ORSI-SOD. DAFNet [19] published the EORSSD dataset, an extension of the ORSSD dataset. The EORSSD dataset addressed more diverse and challenging scenarios compared to the ORSSD dataset. This article proposed a global context-aware attention module that captures the long-range semantic flow among spatial information of multilayer features in an attentive mechanism from the low-level layer to the high-level layer. This article introduced salient edge supervision of the predicted mask additionally. EMFINet [23] designed a novel architecture that utilizes multiscale feature extraction and salient edge prediction. This model integrates the multiscale context features from multiple-resolution input images with edge features extracted from the edge prediction module. MJRBM [37] published a new extensive ORSI-SOD dataset, ORSI4199. ORSI4199 is more extensive and challenging than the two previously published datasets, ORSSD, and EORSSD. MJRBM proposed a joint region and boundary module that fuses semantic features with boundary features effectively. It is based on gate convolution methods like GateNet [12] in NSI-SOD but optimized for ORSI-SOD by utilizing gate information to regions and boundaries of the targets. MCCNet [20] proposed the module that aggregates foreground, background, and edge features in a parallel manner. This model also utilizes the edge module to predict salient edge information. Each feature map extracted in parallel is fused through a module composed of CA and SA. ACCoNet [21] proposed a new architecture that effectively utilizes the interaction of the adjacent features and designed a bifurcation decoder. It shares similarities with LVNet [18] in densely connecting adjacent features like nested structures, but it applies several dilated convolutions that have different receptive fields to ensure the diverse scale attribute of the targets. AESINet [22] applies different modules based on different characteristics between high-level and low-level features. This article proposed an adaptive edge-aware module that utilizes the differential map obtained by subtracting max-pooled features from average-pooled features. To capture SCs more precisely from high-level features, this article proposed a graph convolution network that utilizes node-based information via node-attention mechanisms, and multiscale feature extraction using dilated convolutions was employed. ADSTNet [44] proposes an adaptive dual-stream sparse transformer that combines CNN with a transformer to address the compensation issue between local and global information. MCPNet [45] proposes cross-image dual memory to utilize intrascene knowledge and to discriminate foreground and background more comprehensively.

Many studies have addressed the challenging aspects of ORSI-SOD, such as irregular distributions, diverse shapes and sizes, and even scenarios without targets. However, the difficult aspects of ORSI-SOD have not been adequately addressed yet. In this article, we propose several modules to address the challenging aspects of ORSI-SOD. In Section III, we provide a detailed explanation of our proposed approach.

III. Proposed Method

A. Overview of the Proposed Architecture

The overall pipeline of the proposed architecture is shown in Fig. 2. Our model employs two encoders to extract image context and structure features, respectively. We use Res2Net50 [38] for the image encoder and ResNet18 [39] for the structure encoder. The encoders consist of five layers, each extracting multiscale features of five different sizes. In other words, the multiscale features mean features from five different levels.

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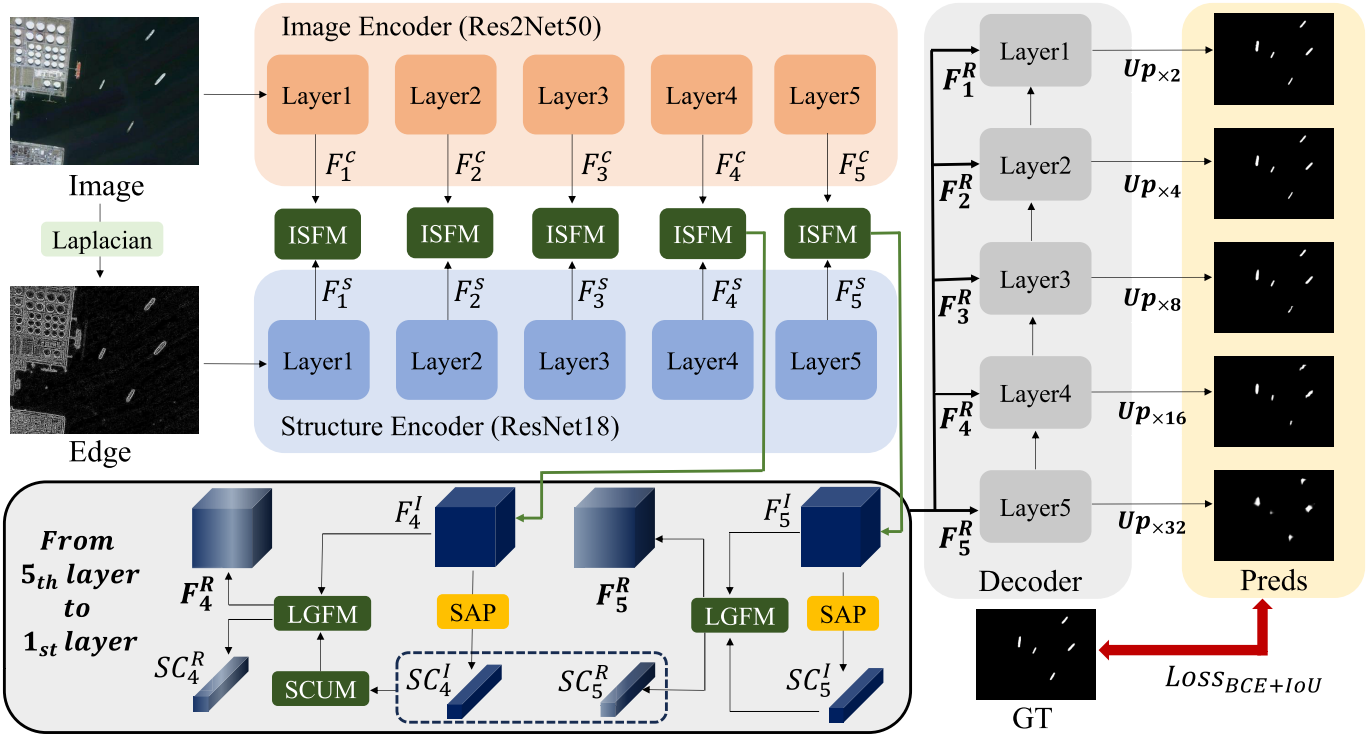


Fig. 2. Overview of our proposed architecture, LSHNet. The RGB image is extracted through the Res2Net encoder to produce context features, $\{F_m^c\}_{m=1}^5$. The edge obtained by passing the RGB image through the Laplacian edge detector is processed through ResNet18 as structure-prior information to produce structure features, $\{F_m^s\}_{m=1}^5$. These two feature maps are fused at each layer through ISFM. The fused features $\{F_m^I\}_{m=1}^5$ are refined into through LFGM and SCUM. LFGM and SCUM are illustrated for the fusion process of the fourth and fifth layers. SAP denotes semantic-aware pooling as illustrated in Fig. 3(a). Similarly, adjacent layers from the fifth to the first layer are fused in the same manner. The refined features $\{F_m^R\}_{m=1}^5$ obtained from the five layers are fed into the decoder, resulting in five outputs. Finally, the model is trained using BCE and IoU loss.

Multiscale features extracted from the two encoders are fused at each level through ISFM. ISFM consists of internal-patch attention and global-patch attention divided by window. ISFM allows for the efficient interpretation of local and global information.

Next, semantic-aware pooling (SAP) is applied to each level of features to extract global features representative of the target. SAP extracts spatially important regions and multiplies with the features before pooing. This method retains only the features around the target center coarsely, representing the distribution of features that characterize the target. We denote these features as SCs. LGFM applies an attention mechanism to the SCs with $c \times 1 \times 1$, using them as keys and values, while using context features from ISFM with $c \times h \times w$ as queries. SCUM deploys from the high-level SCs to low-level SCs. This module refines and fuses two SCs extracted from two adjacent levels by applying gate convolution to each and concatenating them for further refinement. This module integrates SCs from high-level to low-level. In the decoder, we follow a simple U-Net [40] structure, where multilevel features are fused from high-level to low-level. We predict the outputs at each level, resulting in a total of 5 outputs.

In the following Sections III-B–III-E, we provide detailed explanations of the proposed modules.

B. Encoder

Previous studies in ORSI-SOD [19], [20], [23] have focused on predicting only the edges of masks of the targets using additional learnable modules. They did not utilize edge information extracted from images as structure-prior. We aimed to complement the lost structural information during the downsampling of images through the extraction of deep layers and to leverage structural information to achieve a clearer understanding of the scene. We utilize two encoders to extract the features from the image and edge. Res2Net50 [38] and ResNet18 [39] employ image and edge for each. The reason for using a lightweight encoder for edge information is based on the assumption that there is less information in edge and less complexity compared to RGB images, making it possible to extract sufficient information with a smaller network. Structure-prior is obtained by converting the input image to grayscale and applying the Laplacian edge equation as below

$$\text{edge} : \nabla^2 I(x, y) = \frac{\partial^2 I}{\partial^2 x} + \frac{\partial^2 I}{\partial^2 y} \tag{1}$$

$$\text{Kernel} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \tag{2}$$

where x and y denote the coordinate of the image I and Kernel refers to the scope of edge detection in (1). Then, we deploy the two inputs, I and edge, to the encoders in parallel as below

$$\left\{F_m^c\right\}_{m=1}^5 = \text{Enc}_1\left(\text{Input}_{\text{image}}\right) \tag{3}$$

$$\left\{F_m^s\right\}_{m=1}^5 = \text{Enc}_2\left(\text{Input}_{\text{edge}}\right). \tag{4}$$

We obtain deep multiscale features from each encoder as (3) and (4). We denote context features and structure features

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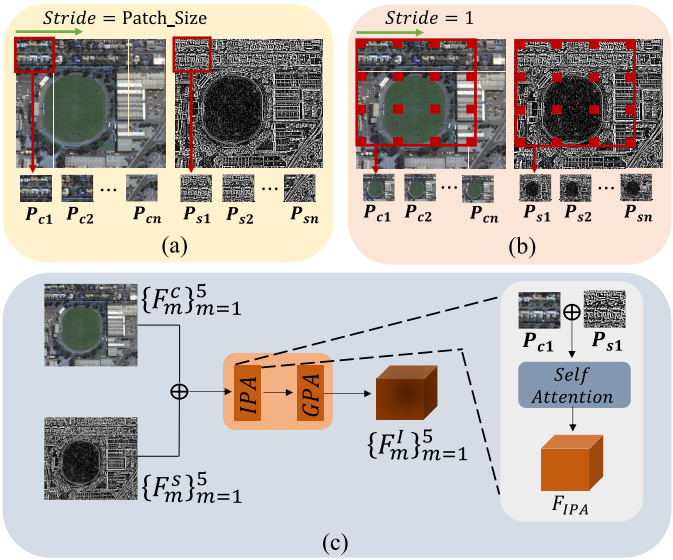


Fig. 3. ISFM. (a) Internal patch. (b) Global patch. (c) Pipeline of ISFM. This module fuses two feature maps in the same layer. As shown in (c), ISFM consists of IPA and GPA. The difference between the two processes lies in the patch sampling method, with (a) representing the internal patch method and (b) representing the global patch method.

as $\{F_m^c\}_{m=1}^5$, $\{F_m^s\}_{m=1}^5$, respectively. m stands for the level of features, and higher numbers indicate higher level features. We embed the number of channels for m from 1 to 5 as $\{64, 128, 256, 512, 512\}$.

C. Image-Structure Fusion Module

ISFM fuses context features extracted from the image encoder and structure features extracted from the structure encoder. Inspired by OmniSR [41], we propose a module consisting of global-patch attention and internal-patch attention to effectively integrate context features with structure features. We define a window size and partition the input based on the window size as below

$$\begin{aligned}\{p_{c1}, p_{c2}, \ldots, p_{cn}\} &= \text{window split}(F_m^c)\\ \{p_{s1}, p_{s2}, \ldots, p_{sn}\} &= \text{window split}(F_m^s)\\ n &= \left\lceil \frac{H}{ws} \right\rceil \times \left\lceil \frac{W}{ws} \right\rceil, \quad \text{ws} = \text{window size}.\end{aligned} \tag{5}$$

In (5), the context features F_m^c and structure features F_m^s are divided into n patches each. We denote patches of the context features and the structure features as p_{cn} and p_{sn}. H and W denote the height and width of the input feature map, respectively. The number of patches n is determined by H, W, and the window size, ws. We use two types of patch sampling methods: internal patch and global patch. Although we illustrated the patch partitioning process on a 2-D image for clarity in Fig. 3, it is actually applied on the feature map, F_m^c, and F_m^s. As shown in Fig. 3(a), we partition the context features and structure features into local patches based on the window size, respectively. As shown in Fig. 3(b), we perform global sampling through windows to capture the overall information of the scene. This sampling divides the entire features into local patches based on the window size and extracts one pixel from each patch to create comprehensive features. While it is similar to a downsampling effect, it can generate n downsampled features without losing the information of input features. Next, we fuse the context patches and structure patches obtained through two types of patch sampling using an attention mechanism. The equations are as below

$$Q, K, V = \text{Conv}(p_{ct} \oplus p_{st}), \quad t = 1, \ldots, n. \tag{6}$$

We sum sampled patches of the context features and structure features. The combined features are embedded as query, key, and values (denoted as Q, K, and V, respectively) as (6)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{7}$$

Then, we implement the attention mechanism following [42]. d_k represents the channel dimension of Q, K, and V

$$\text{FeedForward} = 2 \cdot [\text{Conv}_{1\times1}(\text{GeLU}(\text{Dp}(0.1)))]. \tag{8}$$

The feedforward network involves passing through a 1×1 convolution, followed by GELU activation and dropout as (8). The fusion process can be summarized in the following equation:

$$F_m^I = \text{FeedForward}(\text{Attention}(Q, K, V)). \tag{9}$$

We refer to the application of (9) to Q, K, and V from internal patches as internal patch attention (IPA) and from global patches as global patch attention (GPA) in Fig. 3(c). Through IPA, these n local patches, each with the shape of $[B, C, \text{ws}, \text{ws}]$, are internally processed with attention and then reassembled to produce an output in the shape of $[B, C, H, W]$. This process restricts the receptive fields to the window size, allowing for precise interpretation of local information. In contrast, GPA utilizes n global patches to enhance features related to critical parts of the scene by leveraging comprehensive information. The proposed ISFM consists of two sequential processes: IPA and GPA. We utilize five blocks of this module in the overall structure in Fig. 2 as below

$$\begin{aligned}F_1^I &= \text{ISFM}_{\text{ws}=8}(F_1^c, F_1^s)\\ F_2^I &= \text{ISFM}_{\text{ws}=8}(F_2^c, F_2^s)\\ F_3^I &= \text{ISFM}_{\text{ws}=4}(F_3^c, F_3^s)\\ F_4^I &= \text{ISFM}_{\text{ws}=1}(F_4^c, F_4^s)\\ F_5^I &= \text{ISFM}_{\text{ws}=1}(F_5^c, F_5^s).\end{aligned} \tag{10}$$

We fuse F_m^c and F_m^s extracted from five layers of the two encoders at each corresponding layer from $m = 1$ to $m = 5$ as (10). The window size ws is a hyperparameter specified differently for each layer, with values set to $[8, 8, 4, 1, 1]$ from $m = 1$ to $m = 5$, respectively. As each layer's feature maps have different resolutions, the window size is separately specified for each layer. The fused features $\{F_m^I\}_{m=1}^5$ are refined through LGFM and SCUM to purify the saliency region.

D. Local–Global Feature Fusion Module

LGFM specializes in extracting features specific to the target object while refining the feature maps $\{F_m^I\}_{m=1}^5$ extracted from the previous stage. The overall flow of LGFM is depicted

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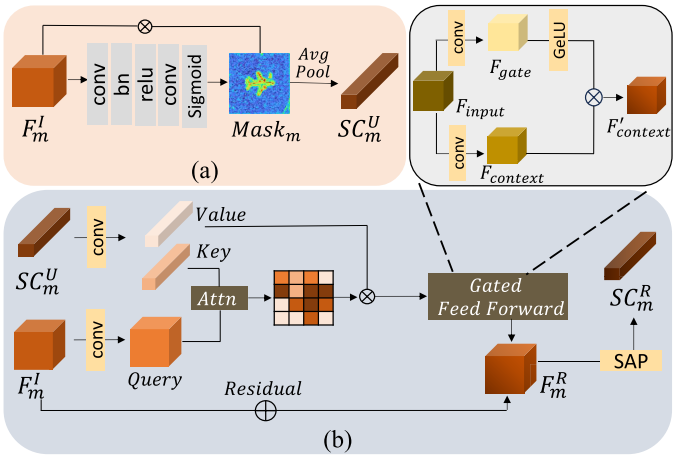


Fig. 4. LGFM. (a) SAP. (b) Pipeline of LGFM. F_{input} is extracted to SCs representing the target through SAP. The SCs are embedded as key and value, and F_{input}, as the query, is matched to the key of the SCs to refine the feature map according to the value. The refined feature map F_m^R produces refined SCs SC_m^R through SAP.

in Fig. 4. As shown in Fig. 4(a), we implement SAP. We first predict the semantic region of the feature maps $\{F_m^I\}_{m=1}^5$, respectively, as below

$$\text{Mask}_m = \text{Sig}\big(\text{Conv}\big(\text{Conv}\big(\text{BN}\big(\text{ReLU}\big(F_m^I\big)\big)\big)\big)\big) \tag{11}$$

Mask approximately predicts the semantic region of the targets in the feature map F_m^I as (11). m represents the feature level, ranging from 1 to 5. This light network employs a single block consisting of a $\text{Conv}_{1\times1}$, batch normalization (BN), and ReLU, followed by applying $\text{Conv}_{1\times1}$ and Sigmoid function. Next, we apply SAP using the obtained mask and feature maps as below

$$\big\{\text{SC}_m^I\big\}_{m=1}^5 = \text{SAP}\Big(\big\{F_m^I\big\}_{m=1}^5, \{\text{Mask}_m\}_{m=1}^5\Big). \tag{12}$$

We multiply the mask of shape $[B, 1, H_m, W_m]$ with the feature map F_m^I of shape $[B, C, H_m, W_m]$, and then perform pooling to generate representative features of the targets with the shape of $[B, C, 1, 1]$. We denote these target-representative features as SCs. This process is performed across five layers, resulting in the creation of five SCs $\{\text{SC}_m^I\}_{m=1}^5$. Since the extracted SCs represent distributions of the target without spatial information, we propose LGFM to convey target-attentive information to the feature map F_m^I, which contains both spatial and channel information of the scene. As shown in Fig. 4(b), we embed the SCs and the feature map as below

$$\text{Key, Value} = \text{Conv}(\text{SC}_m^U), \quad \text{Query} = \text{Conv}\big(F_m^I\big). \tag{13}$$

We set SC SC_m^U as key, value, and F_m^I as query to perform attention. Attention mechanism is same as (7). SC_m^U is an updated SC through SCUM explained in the following Section III-E, hence it is denoted as SC_m^U in (13). In the case of $m = 5$, it corresponds to SC_m^I. This process is illustrated in the interaction between LGFM and SCUM in Fig. 2. The general self-attention mechanism generates attention maps of shape $[B, HW, HW]$ from feature maps of shape $[B, C, H, W]$. However, in this case, SCs (denoted as SC) have only channel information of the targets, resulting in the creation of attention maps of shape $[B, HW, 1]$. Thus, only the regions in HW

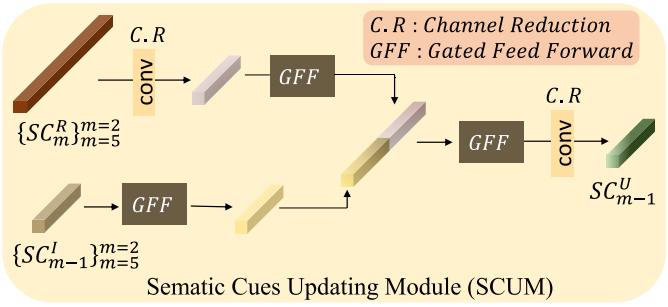


Fig. 5. SCUM. The refined SCs at higher levels SC_m^R are fused with adjacent lower level SCs SC_{m-1}^I for concatenating channel-wise manner. GFF refers to the gated feedforward mechanism as shown in Fig. 4(b).

space of the query that possess features similar to SC_m^U are emphasized, enabling the extraction of SCs-centric feature maps. The obtained features from attention are processed by gated feedforward (GFF). GFF divides the feature maps into two parts: extracting features to be used as gates and extracting context features

$$F_{\text{gate}} = \text{GeLU}(\text{Conv}(F_{\text{input}})). \tag{14}$$

One part of the process, gates utilize the GeLU function to retain important parts of spatial and channel direction information as (14)

$$F_{\text{context}} = \text{Conv}(F_{\text{input}}). \tag{15}$$

The other part, the context part interprets the fused feature maps through the attention mechanism. Integration between two features is as follows:

$$F'_{\text{context}} = F_{\text{gate}} \otimes F_{\text{context}}. \tag{16}$$

These features obtained from (14), (15) are multiplied with element-wise manner in (16). We adopt this mechanism by Restormer [42]. Unlike previous ORSI-SOD studies [20], [21], [22], which separately handle SA and CA, this approach is effective as it can handle them simultaneously. In conclusion, we summarize LFGM as below

$$F_m^R = \text{GFF}(\text{Attention}(Q, K, V)). \tag{17}$$

It applies GFF to the results obtained from (13) to extract the refined context feature maps F_m^R. Attention is the same as (7)

$$\text{SC}_m^R = \text{SAP}\big(F_m^R, \text{Mask}_m^R\big). \tag{18}$$

Then, F_m^R undergoes SAP to output refined representative target features SC_m^R. SAP is the same process as (11) and (12). LGFM produces two outputs. Through this stage, $\{F_m^R\}_{m=1}^5$ for five layers are obtained, and these feature maps are passed to the decoder.

E. SCs Updating Module

SCUM is inserted between the LGFM blocks of each layer and plays a role in integrating information from adjacent layers. At the bottom of Fig. 2, it shows where the SCUM fits into the overall structure. It is shown that before applying LGFM to the fourth layer, SC_5^R from the fifth layer is fused with SC_4^I, and then LGFM is applied to SC_4^U and F_4^I. This process progresses from higher level layers to lower level layers. As shown in Fig. 5, the flow of the module is illustrated.

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SCUM consists of three GFF blocks which are the same as LGFM. We represent the formulas as follows:

$$\mathrm{SC}_{m-1}^{U} = \mathrm{GFF}\left[\mathrm{GFF}\left(\mathrm{CR}\left(\mathrm{SC}_{m}^{R}\right)\right) \parallel \mathrm{GFF}\left(\mathrm{SC}_{m-1}^{I}\right)\right]. \quad (19)$$

As depicted in (19), since higher level SCs SC_m^R have a larger number of channels, the channels are reduced to match the lower level channels corresponding to SC_{m-1}^I in Fig. 5. m represents the layer level and takes values from 2 to 5. $\parallel$ means channel-wise concatenation. Next, both SCs, $\mathrm{SC}_m^{R'}$ and $\mathrm{SC}_{m-1}^{I'}$ pass through GFFs, followed by concatenation in the channel direction, and then pass through GFFs again. This simple architecture enables the SCs to be refined while transmitting information from SCs of adjacent layers. Our approach merges high-level SCs into low-level SCs. High-level features contribute to semantic richness and contextual awareness by large receptive fields, while low-level features provide fine-grained details such as edges and textures by small receptive fields. High-level features bring global context and semantic richness to the local details captured by low-level features. This fusion allows the model to better understand the scene and the relationships between objects. If the target is small or there are multiple targets, the high-level features may be unclear due to their low resolution, but this can be compensated for by the low-level features. We experimentally confirmed that the combination of LGFM and SCUM produces a good synergy, which we address in Section IV-E.

F. Decoder

The decoder followed the U-Net [40] decoder structure, following previous studies [19], [22], [23]. The decoder consists of five layers, with feature maps passed as skip connections to each layer. The feature maps passed as skip connections are $\{F_m^R\}_{m=1}^5$ obtained through LGFM, as shown in Fig. 2. The details of the decoder are as follows:

$$\mathrm{Layer}_m = 2 \cdot [\mathrm{Conv}_{3\times3}, \mathrm{BN}, \mathrm{ReLU}]. \quad (20)$$

The decoder layers are composed of two blocks, each consisting of a convolution (Conv), BN, and ReLU activation as (20)

$$\begin{cases} F_m^d = \mathrm{Layer}_m\left(F_m^R\right), & m = 5 \\ F_m^d = \mathrm{Layer}_m\left(\mathrm{Cat}\left(F_m^R, F_{m+1}^d\right)\right), & m = 1, 2, 3, 4. \end{cases} \quad (21)$$

Except for the last layer, the decoder concatenates the output of the previous layer F_m^d with the input of the current layer F_m^R along the channel dimension to extract features as (21)

$$\begin{cases} S_m = \mathrm{Sigmoid}(\mathrm{Conv}\left(F_m^d\right)), & m = 1, 2, 3, 4, 5 \\ P_m = \mathrm{Up}_m(S_m), & m = 1, 2, 3, 4, 5. \end{cases} \quad (22)$$

Finally, the five features extracted from the decoder $\{F_m^d\}_{m=1}^5$ are used to predict binary masks through Conv and Sigmoid functions, and Each output is then upsampled to match the size of the GT mask for supervised learning as (22). This is a common strategy in the SOD field [11], [12], [20], [21] to enhance the robustness of objects of various sizes.

G. Losses

We use two loss functions: binary cross-entropy (BCE) and intersection over union (IoU). These two losses are widely used not only in ORSI-SOD but also in the segmentation field. The equations of loss functions are as follows (23) and (24), as shown at the bottom of the next page.

H and W denote the height and width of the binary mask, respectively, and i and j represent the coordinate positions. G represents the GT, and P represents the prediction mask. The value of P in the mask represents a probability between 0 and 1. The total loss is represented below

$$\mathcal{L}\mathrm{oss}_{\mathrm{total}} = \sum_{m=1}^{5} \left(\mathcal{L}\mathrm{oss}_{\mathrm{BCE}}(G, P_m) + \mathcal{L}\mathrm{oss}_{\mathrm{IOU}}(G, P_m)\right). \quad (25)$$

We set the weights of the two losses equally to 1 without assigning separate weights. We apply the loss to all five outputs $\{P_m\}_{m=1}^5$. Hence, the total loss function can be expressed as (25).

IV. Experiments

In this section, we provide an overview of the benchmark datasets, details of the training strategy, and an explanation of the evaluation employed in Sections IV-A–IV-C. Subsequently, we delve into the analysis of the experimental results in Section IV-D.

A. Datasets

We conduct experiments on three benchmark datasets.

1) ORSSD (2019): ORSSD dataset [18] consists of 800 ORSIs along with their densely annotated masks. We utilize 600 images for training data and 200 images for test data following [18].

2) EORSSD (2021): EORSSD [19] is an extended version of the ORSSD dataset, consisting of a total of 2000 images and 2000 densely annotated masks. This dataset includes the ORSSD dataset within it. This dataset contains more complicated scenes, challenging object types, and more comprehensive real-world circumstances. We use 1400 images for training data and 600 images for test data following [19].

3) ORSI-4199 (2022): ORSI-4199 dataset [37] is the most recently published dataset among the three public datasets, providing a total of 4199 images and 4199 finely segmented masks. The dataset classifies nine challenging scenarios and provides images classified accordingly. This dataset is the most challenging dataset in ORSI-SOD. We use 2000 images for training data and 2199 images for test data following [37].

B. Evaluation Metrics

Following previous ORSI-SOD works [20], [21], [43], we adopt five metrics for validating our work.

1) S-Measure [46]: S-Measure (S_α) is an evaluation metric that reflects two types of structural similarity: region-aware and object-aware. The weight for ratio α is set to 0.5.

2) F-Measure [47]: F-measure (F_β) is the harmonic mean of precision and recall. β is set to 0.3. This metric is evaluated in three cases: mean, max, and adaptive.

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3) E-Measure [48]: E-measure (E_ξ) is an evaluation metric that simultaneously considers local match at the pixel level and global match at the entire mask level. This metric can be evaluated in three cases: mean, max, and adaptive.

4) Mean Absolute Error (MAE): MAE is a metric used to measure the difference between predicted values and GT values. It is calculated by taking the average of the absolute differences between the predicted and GT values.

5) PR-Curve: PR-Curve is a graphical representation used to evaluate the performance of a classification model. It visualizes the tradeoff between precision and recall. We set the threshold values as 0–255.

C. Implementation Details

We conduct training using the PyTorch framework on an NVIDIA RTX 3090 GPU. The input size is uniformly resized to 320 × 320 during both training and testing. During testing, the output generated at the size of 320 is upsampled to the original size for validation. The output extracted from the first layer of the decoder S_1 in (22) with the highest resolution among the five outputs is used as the final output for validation.

We train on the ORSSD dataset with 600 images using a batch size of 4 for 480 epochs. We employ the Adam optimizer with an initial learning rate of 1e-4. We utilized the MultiStepLR scheduler to halve the learning rate every 120 epochs. We train on the EORSSD dataset with 1400 images using a batch size of 8 for 480 epochs. We utilize the AdamW optimizer with an initial learning rate of 5e-5, halving it every 240 epochs. We train on the ORSI-4199 dataset with 2000 original images augmented using 7 additional augmentation techniques following [18], [22], resulting in a total of 16 000 training images. We set the batch size to 8 and trained for 60 epochs. We used the AdamW optimizer with an initial learning rate of 5e-5, halving it every 30 epochs.

D. Comparison With State-of-the-Art

1) Quantitative Results: We extensively compared the performance across three categories: traditional natural scene SOD methods, NSI-SOD methods, and ORSI-SOD methods. The quantitative results are presented in Table I for ORSSD and EORSSD, and in Table II for ORSI-4199. We compared our model with others on the EORSSD and ORSSD dataset using 8 evaluation metrics following recent works [20], [21], [43]. The performance listed for the EORSSD and ORSSD datasets is only provided for cases using the ResNet backbone. We compared the summarized results using four representative evaluation metrics according to recent works [22], [43]. For ORSI-4199, since it is a recently published dataset, we limited the comparison to the other models that were experimented on this dataset. Therefore, the comparison models listed for EORSSD and ORSSD differ from those listed for ORSI-4199.

NSI-SOD methods show a significant performance gap compared to ORSI-SOD methods. Since NSI-SOD is optimized for objects in natural scenes, it performs poorly in ORSI-SOD. The target types in ORSI-SOD exhibit much more diverse and complex patterns compared to those in NSI-SOD. This highlights the necessity for further research in ORSI-SOD. Although our proposed model does not achieve the best performance in all metrics, it demonstrates the best performance in 6 out of 8 metrics for ORSSD and in 4 out of 8 metrics for EORSSD, and in 3 out of 4 metrics for ORSI-4199 as shown in Tables I and II (marked in red block). Depicted in Tables I and II, our model demonstrates outstanding overall performance, substantiating our excellent generalization capabilities. Comparing specific metrics in ORSSD to analyze more detail, our model achieves 0.9491 of S_α, surpassing the second-best by 0.0036 points. This difference is quite a significant improvement, exceeding the gap between the second and third-best performance, which is 0.0018 points, by twice the amount. In $F_\beta^{\max}$ of ORSSD, we achieve a value of 0.9200, which is an improvement of 0.0040 over the second-best result. The difference between the second-best and third-best results is only 0.0005. This improvement is quite substantial when compared to other studies. In the MAE metric, we also achieve superior performance with a value of 0.0075, outperforming other models. As shown in Fig. 6, our model is positioned at the top of the PR curve, demonstrating an excellent balance between precision and recall.

In EORSSD, our model and CRNet [43] achieve the same performance in S_α metric. However, when considering all 8 metrics, our model outperforms CRNet in 5 of them, and the difference in MAE is a mere 0.0001, indicating a very slight margin. Our model shows significant improvements over other models in both F_β^{adp} and E_ξ^{adp} metrics. For F_β^{adp}, it achieves a score of 0.8261, surpassing the second-best by 0.0121 points, while the difference between the second-best and third-best is only 0.0003. For E_ξ^{adp}, our model achieves 0.9619, surpassing the second-best by 0.0056 points, while the difference between the second-best and third-best is 0.0025. Adaptive thresholding denoted as adp is a method where the threshold for a binary mask is determined by choosing the smaller value between the mean of the predicted distribution or fixed maximum value, aiming to reflect the relativeness of predicted distributions more precisely. DAFNet [19] shows the best performance in the MAE metric but performs poorly in the other metrics. This indicates a tradeoff between other metrics, making it challenging to achieve superior results across all metrics simultaneously. As shown in Fig. 6, our model is positioned at the top in terms of both precision and recall. From this

$$\mathcal{L}\text{oss}_{\text{BCE}} = \sum_{i=1}^{H}\sum_{j=1}^{W} -\left[G(i,j)\log(P(i,j)) + (1-G(i,j))\log(1-P(i,j))\right] \tag{23}$$

$$\mathcal{L}\text{oss}_{\text{IOU}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}(G(i,j)\cdot P(i,j))}{\sum_{i=1}^{H}\sum_{j=1}^{W}\left[G(i,j)+P(i,j)-G(i,j)\cdot P(i,j)\right]} \tag{24}$$

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TABLE I

QUANTITATIVE RESULTS ON EORSSD, ORSSD. TYPE REFERS TO THE TYPE OF METHOD: T.N MEANS TRADITIONAL NATURAL SCENE IMAGE METHOD, C-NSI MEANS CONVOLUTION-BASED NATURAL SCENE IMAGE METHODS, AND C-RSI MEANS CONVOLUTION-BASED REMOTE SENSING IMAGE METHODS. ↑ MEANS HIGHER IS BETTER AND ↓ MEANS LOWER IS BETTER. BEST RESULTS MARKED IN RED, SECOND-BEST RESULTS IN ORANGE, THIRD-BEST RESULTS MARKED IN YELLOW

Methods	Publication	Type	EORSSD								ORSSD							
			$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
RRWR [33]	*CVPR (2015)*	T.N	0.5992	0.3993	0.3686	0.3344	0.6894	0.5943	0.5639	0.1677	0.6835	0.5590	0.5125	0.4874	0.7649	0.7017	0.6949	0.1324
RCRR [34]	*TIP (2018)*	T.N	0.6007	0.3995	0.3685	0.3347	0.6882	0.5946	0.5636	0.1644	0.6849	0.5591	0.5126	0.4876	0.7651	0.7021	0.6950	0.1277
EGNet [15]	*ICCV (2019)*	C-NSI	0.8601	0.7880	0.6967	0.5379	0.957	0.8775	0.7566	0.0110	0.8721	0.8332	0.7500	0.6452	0.9731	0.9031	0.8226	0.0216
PoolNet [36]	*CVPR (2019)*	C-NSI	0.8207	0.7545	0.6406	0.4611	0.9292	0.8193	0.6836	0.0210	0.8403	0.7706	0.6999	0.6166	0.9343	0.8650	0.8124	0.0358
MINet [11]	*CVPR (2020)*	C-NSI	0.9040	0.8344	0.8174	0.7705	0.9442	0.9346	0.9243	0.0093	0.9040	0.8761	0.8574	0.8251	0.9545	0.9454	0.9423	0.0144
GateNet [12]	*ECCV (2020)*	C-NSI	0.9114	0.8566	0.8228	0.7109	0.9610	0.9385	0.8909	0.0095	0.9186	0.8871	0.8679	0.8229	0.9664	0.9538	0.9428	0.0137
U2Net [13]	*PR (2020)*	C-NSI	0.9199	0.8732	0.8329	0.7221	0.9649	0.9373	0.8989	0.0076	0.9162	0.8738	0.8492	0.8038	0.9539	0.9387	0.9326	0.0166
SUCA [17]	*TMM (2021)*	C-NSI	0.8988	0.8229	0.7949	0.7260	0.9520	0.9277	0.9082	0.0097	0.8989	0.8484	0.8237	0.7748	0.9584	0.9400	0.9194	0.0145
PA-KRN [14]	*AAAI (2021)*	C-NSI	0.9192	0.8639	0.8358	0.7993	0.9616	0.9536	0.9416	0.0104	0.9239	0.8890	0.8727	0.8548	0.9680	0.9620	0.9579	0.0139
LVNet [18]	*TGRS (2019)*	C-RSI	0.8630	0.7794	0.7328	0.6284	0.9254	0.8801	0.8445	0.0146	0.8815	0.8263	0.7995	0.7506	0.9456	0.9259	0.9195	0.0207
DAFNet [19]	*TIP (2021)*	C-RSI	0.9184	0.8734	0.7980	0.6522	0.9815	0.9382	0.8506	0.0053	0.9188	0.8999	0.8442	0.7565	0.9821	0.9537	0.9115	0.0106
MJRBM [37]	*TGRS (2022)*	C-RSI	0.9197	0.8656	0.8239	0.7066	0.9646	0.9350	0.8897	0.0099	0.9204	0.8842	0.8566	0.8022	0.9623	0.9415	0.9328	0.0163
EMFINet [23]	*TGRS (2022)*	C-RSI	0.9319	0.8742	0.8505	0.8036	0.9712	0.9598	0.9500	0.0075	0.9432	0.9155	0.9000	0.8797	0.9813	0.9726	0.9715	0.0095
MCCNet [20]	*TGRS (2022)*	C-RSI	0.9327	0.8904	0.8604	0.8137	0.9755	0.9685	0.9538	0.0066	0.9437	0.9155	0.9054	0.8957	0.9800	0.9758	0.9735	0.0087
ACCoNet [21]	*TCYB (2023)*	C-RSI	0.9302	0.8821	0.8462	0.7783	0.9759	0.9659	0.9368	0.0067	0.9428	0.9149	0.8952	0.8830	0.9819	0.9776	0.9735	0.0087
ERPNet [49]	*TCYB (2023)*	C-RSI	0.9252	0.8743	0.8269	0.7170	0.9665	0.9366	0.8994	0.0082	0.9352	0.9036	0.8798	0.8392	0.9738	0.9064	0.9521	0.0114
CRNet [43]	*TGRS (2023)*	C-RSI	0.9370	0.8873	0.8653	0.8140	0.9743	0.9618	0.9563	0.0063	0.9389	0.9107	0.8951	0.8695	0.9793	0.9687	0.9711	0.0091
AESINet [22]	*TGRS (2023)*	C-RSI	0.9347	0.8792	0.8496	0.7909	0.9757	0.9647	0.9480	0.0064	0.9455	0.9160	0.8962	0.8640	0.9814	0.9741	0.9689	0.0085
ADSTNet [44]	*JSTARS (2024)*	C-RSI	0.9311	0.8804	0.8716	0.8532	0.9769	0.9709	0.9681	0.0065	0.9379	0.9124	0.9042	0.8979	0.9807	0.9740	0.9785	0.0086
MCPNet [45]	*TGRS (2024)*	C-RSI	0.9373	0.8868	0.8620	0.8077	0.9765	0.9657	0.9515	0.0070	0.9433	0.9135	0.8979	0.8726	0.9807	0.9736	0.9724	0.0090
LSHNet (Ours)	-	**C-RSI**	**0.9370**	**0.8844**	**0.8643**	**0.8261**	**0.9761**	**0.9692**	**0.9619**	**0.0064**	**0.9491**	**0.9200**	**0.9054**	**0.8848**	**0.9824**	**0.9764**	**0.9758**	**0.0075**

TABLE II

QUANTITATIVE RESULTS ON ORSI4199. WE COMPARED OUR MODEL WITH SEVEN STATE-OF-THE-ART ORSI-SOD MODELS. C-RSI MEANS CONVOLUTION-BASED REMOTE SENSING IMAGE METHODS. BEST RESULTS MARKED IN RED, SECOND-BEST RESULTS IN ORANGE, THIRD-BEST RESULTS MARKED IN YELLOW

Methods	Publication	Type	ORSI-4199			
			$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$
MJRBM [37]	*TGRS (2022)*	C-RSI	0.8582	0.8511	0.9343	0.0372
EMFINet [23]	*TGRS (2022)*	C-RSI	0.8712	0.8636	0.9403	0.0313
MCCNet [20]	*TGRS (2022)*	C-RSI	0.8746	0.8690	0.9413	0.0316
ACCoNet [21]	*TCYB (2023)*	C-RSI	0.8805	0.8688	0.9424	0.0320
ERPNet [49]	*TCYB (2023)*	C-RSI	0.8636	0.8528	0.9292	0.0388
CRNet [43]	*TGRS (2023)*	C-RSI	0.8669	0.8669	0.9176	0.0335
AESINet [22]	*TGRS (2023)*	C-RSI	0.8755	0.8726	0.9459	0.0305
ADSTNet [44]	*JSTARS (2024)*	C-RSI	0.8710	0.8698	0.9433	0.0318
MCPNet [45]	*TGRS (2024)*	C-RSI	0.8736	0.8667	0.9402	0.0324
LSHNet (Ours)		C-RSI	**0.8759**	**0.8758**	**0.9462**	**0.0299**

observation, our model's ability to consistently achieve top-tier results across all metrics demonstrates its superiority.

As shown in Table II, our model significantly outperforms the performance of MJRBM [37], the article that published ORSI-4199 dataset, across all four metrics. MCCNet [20] demonstrates competitive performance with our model in both ORSSD and EORSSD. However, it falls behind our model in all metrics in ORSI-4199. In the $F_\beta^{\max}$ metric, both AESINet and our model outperform other models by a significant margin. While AESINet scores 0.0036 points higher than MCCNet, our model demonstrates even superior performance with a 0.0068-point lead over MCCNet. ACCoNet [21] achieves a score of 0.8805 in S_α metric, showing the highest performance. However, it shows relatively poor results in other metrics. This illustrates that focusing too much on one metric performance may lead to a decline in the performance of different metrics. In the PR curve of Fig. 6, our model is positioned at the top except for the extreme recall threshold region.

ADSTNet [44] and MCPNet [45], as the most recently published models show impressive performance. ADSTNet outperforms our model on E_ξ and F_β metrics for the EORSSD dataset and MCPNet surpasses our model on the S_α metric for the same dataset. However, on the ORSSD dataset, our model achieves superior results in 5 out of 8 metrics, and on the ORSI-4199 dataset, our model outperforms in 3 out of 4 metrics. ORSI-4199 is considered one of the most challenging datasets recently released. The fact that our model performs well on this dataset demonstrates that our proposed method works effectively even on challenging data, validating its robustness. Also, our model remains competitive on the EORSSD dataset.

2) Qualitative Results: We conduct an extensive visual comparison of qualitative results. Fig. 7 presents the prediction results of ORSSD and EORSSD for traditional methods, NSI-SOD methods, and ORSI-SOD methods. RCRR [34] is represented as a traditional method and is shown in the last

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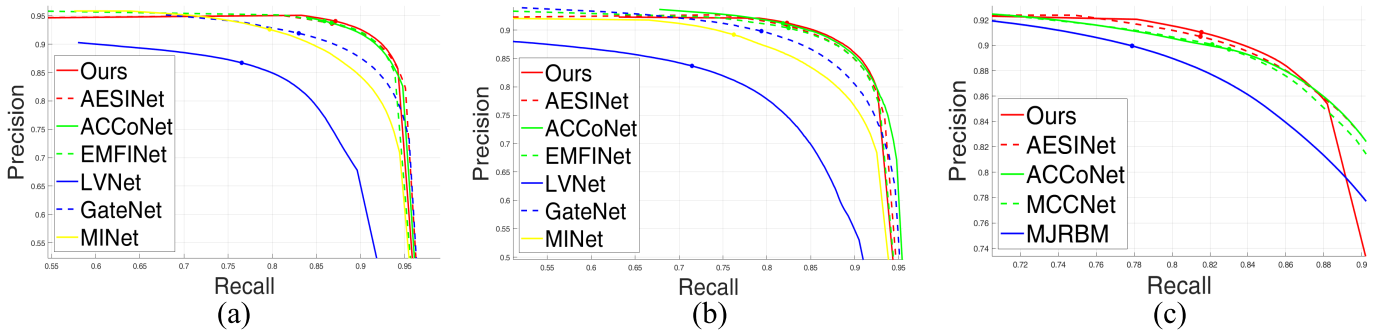


Fig. 6. PR curve on ORSSD, EORSSD, and ORSI-4199. The color of the line and the corresponding model are shown together in the figure. (a) ORSSD. (b) EORSSD. (c) ORSI-4199.

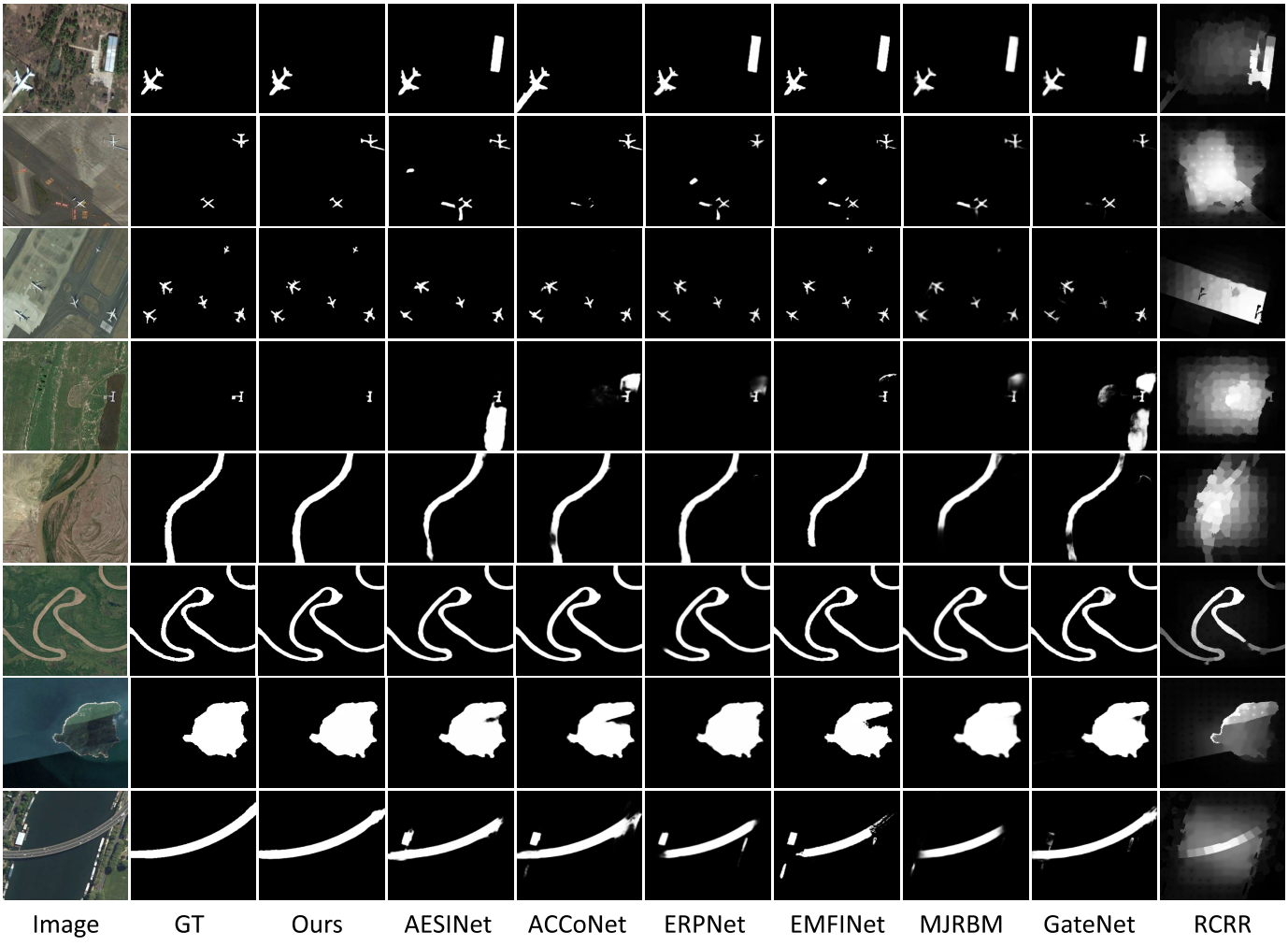


Fig. 7. Qualitative results on ORSSD and EORSSD. We focused on comparing our model with competitive ORSI-SOD methods. GateNet [12] is an NSI-SOD method, RCRR [33] is a traditional method, and the rest are all ORSI-SOD methods (AESINet [22], ACCoNet [21], ERPNet [49], EMFINet [23], and MJRBM [37]). The bottom row of the figure indicates which model's prediction the figure represents.

column of Fig. 7. Traditional methods rely on algorithms determined by differences in pixel values and energy based on distributions to perform segmentation, which may result in poor performance in complex patterns and low-quality images in ORSI-SOD. GateNet [12], a CNN-based NSI-SOD method, is represented in the second-to-last column of the results. While GateNet roughly captures salient objects, when compared to other ORSI-SOD methods, it fails to predict accurately, as seen in the second, third, and fifth rows of the images.

In the first row of images, our model shows results closest to the GT. Most other models incorrectly identify buildings along with the airplane as salient objects. However, since the essence of salient object detection lies in identifying the most prominent single feature, including both airplanes and buildings as salient objects is erroneous, as they possess

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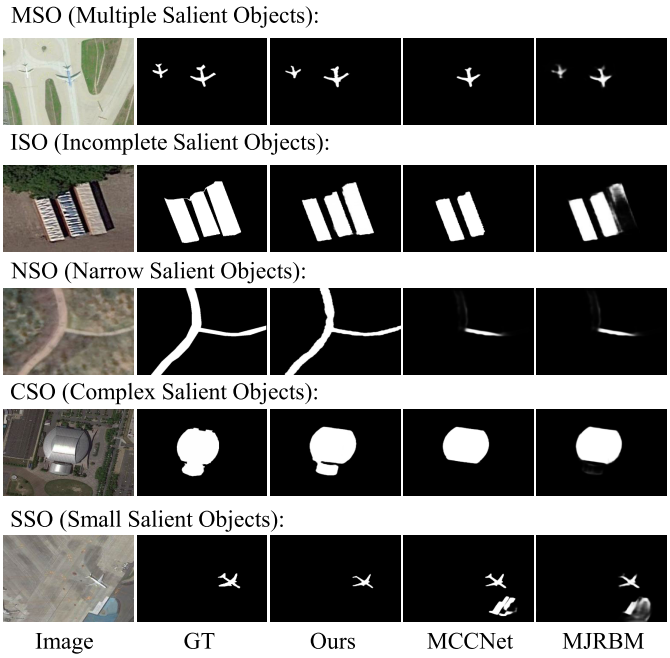


Fig. 8. Qualitative results on ORSI-4199. The sentence above the figure signifies the scenario depicted in the image. The bottom row of the figure indicates which model's prediction the figure represents.

distinct characteristics. In the third row of images, there is a very small airplane in the upper right corner. Most models fail to detect this. While EMFINet and MJRBM successfully detected it, our model achieved more precise segmentation for all five airplanes. The image in the fourth row also features a very small object, making it a challenging case due to the inclusion of terrain that differs from its surroundings. Although our model also fails to achieve perfect segmentation, our model successfully detects only the main object, whereas other models detect the surrounding terrain as well. This demonstrates that our model is better at focusing on the salient object and avoiding the inclusion of irrelevant background, highlighting its superior precision compared to other models. In the fifth row of images, the target terrain is very similar to its surrounding background, making it a challenging case. Our model produces predictions that nearly match the GT, effectively distinguishing the target from the background. However, other models struggle to differentiate the background from the target, resulting in either broken lines or misidentifying the surrounding background as part of the target. This demonstrates our model's superior ability to handle challenging cases with similar foreground and background, ensuring accurate segmentation. In the seventh row of images, the target terrain is shadowed. Our model accurately predicts the target terrain, unaffected by the shadow. In contrast, other models fail to accurately predict the terrain due to the shadow, misidentifying the shadowed areas as background. This demonstrates our model's robustness in handling challenging conditions like shadow.

Fig. 8 presents qualitative results for the ORSI-4199 dataset. This dataset includes nine challenging scenarios, and we have displayed images from five of these scenarios in our results. In the first row, which represents the MSO case, our model successfully detects both airplanes. MCCNet fails to detect the airplane that has a background color similar to it and MJRBM inaccurately predicts the shape of the airplane. In the second row, representing the ISO case, the target is partially obscured by the surrounding terrain. Our model cleanly separates the target building from the surrounding trees. In contrast, other models fail to identify the building as a salient object due to it being obscured by the terrain, demonstrating our model's superior ability to distinguish targets even in challenging conditions. In the third row, representing the NSO case, the target terrain has a complex and thin shape. Our model predicts the target with high similarity to the GT, while other models fail to accurately detect the target, demonstrating the superior precision of our model in handling intricate and fine structures. In the CSO case, our model produces predictions most similar to the GT compared to other models, which fail to detect various parts of the building altogether. In the SSO case, other models incorrectly identify the surrounding background as the main object, whereas our model successfully detects only the airplane. This highlights the superior accuracy and reliability of our model in distinguishing salient objects from complex backgrounds.

TABLE III

ABLATION STUDIES ON OUR PROPOSED MODULES. B REPRESENTS THE BASE MODEL. ED STANDS FOR EDGE ENCODER, IS REPRESENTS ISFM, LG DENOTES LGFM, AND SC DENOTES SCUM. DETAILED EXPLANATIONS ARE PROVIDED IN IV-E1

Model	B	ED	IS	LG	SC	ORSSD			
						$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
1	✓					0.9388	0.9063	0.9719	0.0098
2	✓	✓				0.9430	0.9126	0.9777	0.0091
3	✓	✓	✓			0.9462	0.9151	0.9785	0.0085
4	✓	✓	✓	✓		0.9458	0.9151	0.9787	0.0082
5	✓	✓	✓	✓	✓	**0.9491**	**0.9200**	**0.9824**	**0.0075**

Model	B	ED	IS	LG	SC	EORSSD			
						$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
1	✓					0.9326	0.8758	0.9706	0.0067
2	✓	✓				0.9336	0.8801	0.9744	0.0075
3	✓	✓	✓			0.9350	0.8780	0.9715	0.0067
4	✓	✓	✓	✓		0.9352	0.8825	0.9757	0.0064
5	✓	✓	✓	✓	✓	**0.9370**	**0.8844**	**0.9761**	**0.0064**

E. Ablation Studies

We investigate various factors affecting performance. First, we validate the efficiency of our proposed modules. The subsequent sections analyze performance differences due to input size, the impact of different backbone networks, and, finally, a detailed examination of failure cases.

1) Effectiveness of the Proposed Modules: To demonstrate the effectiveness of our proposed module, we conduct experiments on two datasets using five different cases as shown in Table III. The base model (B) consists of connecting the encoder and decoder through skip connections. The edge encoder (ED) is an additional encoder designed to extract edge information more deeply. We sum the structure features

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extracted from the edge encoder with the features extracted from the image encoder and pass them to the decoder. ISFM denoted as IS (III-C) applies the ISFM module to the combined feature maps described earlier. LGFM denoted as LG (III-D) is applied to the feature maps obtained through ISFM. SCUM denoted as SC (III-E) integrates features among SCs extracted from LGFM of the multilayer.

The edge encoder demonstrates improvements across both datasets. S_α, which applies the SSIM metric to the mask, is a metric that enables precise evaluation of the boundary regions of predictions. In particular, the introduction of the edge encoder on the ORSSD dataset resulted in a significant improvement of 0.0042 points in S_α. The improvement in this metric highlights the importance of leveraging the deep structure-prior information from the structure encoder. The effectiveness of ISFM is illustrated in Model 3 in both tables. This module demonstrates improvements across all four metrics in ORSSD. In addition, in EORSSD, it shows promising enhancements in S_α and $\mathcal{M}$. In EORSSD, there is a slight decrease in $F_\beta^{\max}$ and $E_\xi^{\max}$, which is observed to be due to the inclusion of background information during the integration of local and global information by ISFM. However, when considering all metrics together, this module shows improvements across a total of six metrics in two datasets. Particularly in ORSSD, it demonstrates enhancements of 0.0032 points in S_α and 0.0025 points in $F_\beta^{\max}$. As shown in Table I, With just the introduction of the ED and IS modules, our model already demonstrates top-tier results in ORSSD.

We propose LGFM to extract feature maps more focused on salient objects, and the results are presented in Model 4. The effect of LGFM is minimal in ORSSD. However, it shows a significant improvement in all metrics in EORSSD. In EORSSD, LGFM exhibits an improvement of 0.0045 points in $F_\beta^{\max}$ and 0.0042 points in $E_\xi^{\max}$. The results suggest that LGFM is effective, particularly on the relatively challenging dataset EORSSD. SCUM is a module proposed to maximize the effectiveness of LGFM. It shows improvements across all metrics on both datasets. SCUM shows significant improvements in S_α for both ORSSD and EORSSD, with gains of 0.0033 and 0.0018, respectively. In terms of $F_\beta^{\max}$, it demonstrates notable enhancements of 0.0049 for ORSSD and 0.0019 for EORSSD. These results highlight the importance of feature fusion across multiple layers. These results highlight the importance of feature fusion across multiple layers.

Overall, our proposed methods demonstrate significant improvements over the base model. In ORSSD, we observe enhancements of 0.0103 in S_α, 0.0137 in $F_\beta^{\max}$, 0.0105 in $E_\xi^{\max}$, and 0.0023 in $\mathcal{M}$. Similarly, in EORSSD, improvements of 0.0044 in S_α, 0.0086 in $F_\beta^{\max}$, 0.0055 in $E_\xi^{\max}$, and 0.0003 in $\mathcal{M}$ are observed. These results attest to the superiority of the modules we proposed.

2) Qualitative Comparison on LGFM: As shown in Fig. 9, we demonstrate the difference in prediction results with and without LGFM. We select various scenarios for comparison. In the first row, where the target object is thin, the model without LGFM struggles to detect the complete shape, showing discontinuities in the middle. However, with LGFM, the model consistently identifies the irregular shape of the target object.

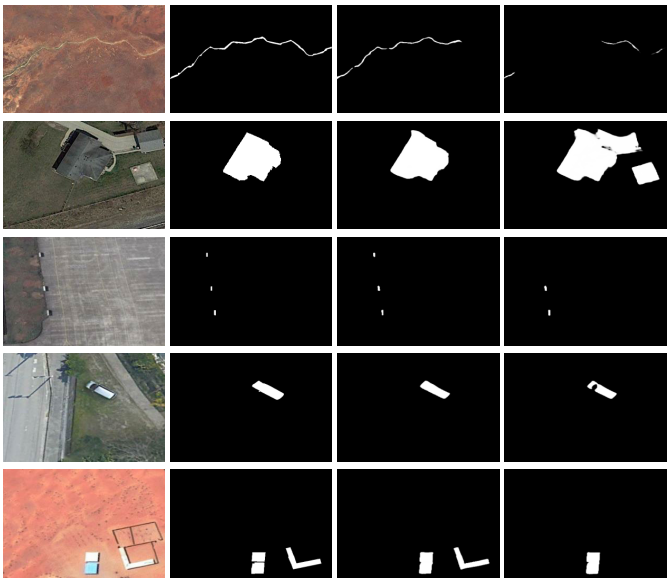


Fig. 9. Qualitative comparison on LGFM. The above results validate the impact of LGFM across various scenarios, including when objects are thin, small, or multiple, or when the target objects have different shapes.

The second row depicts a scenario where the background distractors have a significant influence, and the target object's shape is irregular. Without LGFM, the model incorrectly predicts other areas as part of the target. However, when LGFM is applied, the model successfully separates the target object from the background. The third row shows a scenario where small objects are the target. Without LGFM, the model fails to detect one of the target objects entirely. However, with LGFM, all target objects are successfully captured. The fourth row presents a scenario where the target object has varying colors within itself. Without LGFM, the model fails to recognize the windshield of the car, resulting in a disconnected prediction. However, with LGFM, the model provides a correct prediction that aligns with the overall characteristics of the car. The fifth row illustrates a scenario with multiple target objects of different shapes. Without LGFM, the model completely misses one of the objects with a distinct shape. However, with LGFM, the features are refined to focus on the core characteristics of the target objects, successfully detecting all objects, even those with differing shapes.

3) Analysis on the Updating Methods of SCUM: We propose SCUM, a module for integrating SCs from multilevels. We compare two methods: merging SCs from high-level to low-level and low-level to high-level. As shown in Table IV, the results indicate that the high-level to low-level update strategy consistently outperforms the low-level to high-level approach across all metrics. The difference in $F_\beta^{\max}$ scores is quite pronounced, with the high-level to low-level method achieving an improvement of 0.0032 points on the ORSSD dataset and 0.0033 points on the EORSSD dataset. As shown in Fig. 10, the high-level to low-level update method is more effective in removing background distractors and capturing more accurate regions of the targets. The first and second rows involve scenarios where large objects are detected in the scene. In these cases, the low-to-high update method misjudges

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TABLE IV
COMPARISON ON FEATURE UPDATE METHODS OF SCUM: HIGH-TO-LOW VERSUS LOW-TO-HIGH IN ABLATION STUDIES. THE BEST RESULTS MARKED ON BOLD

Method	ORSSD			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
$Low \rightarrow High$	0.9447	0.9168	0.9795	0.0087
$High \rightarrow Low$ (Ours)	**0.9491**	**0.9200**	**0.9824**	**0.0075**
Method	EORSSD			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
$Low \rightarrow High$	0.9365	0.8811	0.9742	0.0066
$High \rightarrow Low$ (Ours)	**0.9370**	**0.8844**	**0.9761**	**0.0064**

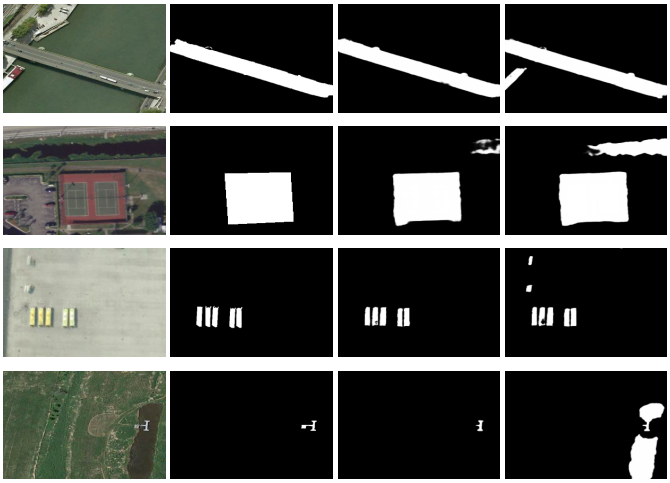


Fig. 10. Qualitative comparison on feature update methods of SCUM. H-to-L refers to the method of updating from high-level to low-level. L-to-H represents the reverse process. H-to-L method demonstrates robustness in removing background distractors.

other regions as salient objects compared to the GT. The third and fourth rows are scenarios where small objects are being detected. Similarly, the low-level to high-level update method incorrectly predicts surrounding regions as salient objects, in addition to the target object. Through these observations, the high-level to low-level approach ensures that the detailed information is guided by more abstract, semantically rich representations, improving the coherence between the high-level and low-level features. These findings support the notion that higher level SCs, which are more specialized in capturing object-specific information, provide more benefits for refining lower level features than the method that updates low-level to high-level.

4) Influence of Input Size: Previous studies have employed various input sizes appropriate for their proposed methods. For instance, MCCNet [20], AESINet [22], and EMFINet [23] use a size of 256 × 256, while DAFNet [19] and GPNet [50] use 128 × 128, CRNet [43] uses 224 × 224, HFANet [51] uses 448 × 448, and MJRBM [37] uses 352 × 352. Accordingly, we conduct our experiments using four input sizes to find the optimal size of our model. As shown in Table V, we empirically observe that input size has a significant impact on performance across different datasets; 224 × 224 size

TABLE V
ABLATION STUDIES ON THE INFLUENCE OF INPUT SIZE. BEST RESULTS MARKED ON BOLD

Input size	ORSSD			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
224 × 224	0.9434	0.9159	**0.9845**	0.0085
256 × 256	0.9432	0.9125	0.9800	0.0088
320 × 320	**0.9491**	0.9200	0.9824	**0.0075**
448 × 448	0.9486	**0.9202**	0.9822	0.0079
Input size	EORSSD			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
224 × 224	0.9334	0.8753	0.9766	0.0066
256 × 256	0.9340	0.8781	0.9756	0.0064
320 × 320	0.9370	0.8844	0.9761	0.0064
448 × 448	**0.9408**	**0.8908**	**0.9781**	**0.0062**
Input size	ORSI-4199			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
224 × 224	0.8762	0.8726	0.9449	0.0311
256 × 256	**0.8778**	**0.8760**	**0.9476**	**0.0290**
320 × 320	0.8759	0.8758	0.9462	0.0299
448 × 448	0.8765	0.8716	0.9431	0.0306

demonstrates relatively weaker performance on all datasets; 256 × 256 size performs best on the ORSI-4199 dataset but shows poor results on ORSSD and EORSSD; 448 × 448 size achieves excellent performance on EORSSD but does not perform well on the ORSI-4199 dataset, except for S_α. These results demonstrate that increasing the input size is not a guaranteed solution to improve performance; 320 × 320 size offers the most balanced performance across all three datasets. While 320 × 320 size may not be the top performer for any single dataset, it provides consistently strong performance across all datasets. Prioritizing generalized performance, we have selected 320 × 320 as the optimal input size. In both the ablation studies and comparisons with other models (shown in Table I, II), 320 × 320 size consistently shows the most balanced results.

5) Selection of Backbones for Image Encoder and Structure Encoder: We conduct experiments to evaluate the impact of different backbones for the image encoder and structure encoder within a dual-branch encoder architecture. The backbones of the image encoder used are VGG16 [52], ResNet50 [39], and Res2Net50 [38], commonly employed in ORSI-SOD tasks. The results of these experiments are presented in Table VI. Our proposed modules are all included and the structure backbone is fixed as ResNet18. We observed that the Res2Net50 backbone consistently outperformed VGG and ResNet50 for the Image Encoder. This is likely due to Res2Net50's ability to capture multiscale features more effectively, which is crucial for processing complex patterns and irregular distributions of objects in ORSIs. Its architecture, which constructs hierarchical residual-like connections within one single residual block, allows for better pattern

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TABLE VI

ABLATION STUDIES ON THE BACKBONES OF THE IMAGE ENCODER. THE BEST RESULTS MARKED ON BOLD

Backbone	ORSSD			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
VGG16	0.9390	0.9086	0.9767	0.0118
ResNet50	0.9440	0.9153	0.9814	0.0082
Res2Net50 (Ours)	**0.9491**	**0.9200**	**0.9824**	**0.0075**

Backbone	EORSSD			
	$S_\alpha \uparrow$	F_β^{max}	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
VGG16	0.9345	0.8783	0.9725	0.0085
ResNet50	0.9361	0.8816	**0.9769**	0.0065
Res2Net50 (Ours)	**0.9370**	**0.8844**	0.9761	**0.0064**

TABLE VII

ABLATION STUDIES ON THE BACKBONES OF THE STRUCTURE ENCODER. THE BEST RESULTS MARKED ON BOLD

Backbone	ORSSD			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
ResNet18 (Ours)	**0.9491**	**0.9200**	**0.9824**	**0.0075**
ResNet34	0.9445	0.9152	0.9775	0.0091
ResNet50	0.9456	0.9143	0.9788	0.0083

Backbone	EORSSD			
	$S_\alpha \uparrow$	F_β^{max}	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
ResNet18 (Ours)	0.9370	0.8844	0.9761	0.0064
ResNet34	0.9370	0.8851	0.9764	**0.0062**
ResNet50	**0.9380**	**0.8869**	**0.9769**	0.0065

recognition at different scales, leading to improved model performance. While VGG and ResNet50 have simpler architectures, they were less efficient in capturing complex patterns in ORSI-SOD, resulting in lower performance compared to Res2Net50. There is a significant difference between the results obtained with the VGG16 and Res2Net50 backbones in ORSSD. Notably, there are a considerable difference in S_α, with 0.0101, and $F_\beta^{\max}$, with 0.0114. In addition, $\mathcal{M}$ shows a difference of 0.0043. Since the size of the ORSSD dataset is smaller, VGG16 might be more prone to overfitting, leading to these results. ResNet50 also shows inferior results compared to Res2Net50. However, it outperforms VGG16 by a large margin. When comparing ResNet50 to Res2Net50, there is a difference of 0.0051 in S_α and 0.0047 in $F_\beta^{\max}$. The differences in $E_\xi^{\max}$ and $\mathcal{M}$ are minimal, at only 0.0010 and 0.0007, respectively. The performance trend continues in EORSSD, where Res2Net50 shows the best performance. In comparison to VGG16 and ResNet50, Res2Net50 outperforms them by 0.0025 and 0.0009 in S_α, and 0.0061 and 0.0028 in $F_\beta^{\max}$, respectively.

Based on the above experimental results, we fix the Image Encoder to Res2Net50 and conduct experiments by varying the backbone of the Structure Encoder. In the experimental results represented in Table VII, a tradeoff is observed depending on the size of the Structure Encoder. ResNet18 achieves the best performance on ORSSD, while ResNet50 performs the best on EORSSD. This is analyzed as being due to the size of the training datasets. ORSSD is a smaller dataset with only 600 training images, whereas EORSSD is a larger dataset consisting of 2000 training images. We find that the performance of the structure encoder, which utilizes edge information, varies depending on the dataset size. On larger datasets, ResNet50 performs better due to its deeper architecture and ability to interpret more complex and diverse scenarios. However, on smaller datasets, ResNet18 shows superior performance, likely because its relatively simpler structure avoids overfitting and makes better use of the limited available data. Specifically, there is a difference of 0.0035 points in S_α and 0.0047 points in $F_\beta^{\max}$ between ResNet18 and ResNet50 on ORSSD. On the other hand, in the EORSSD dataset, ResNet50 outperforms ResNet18 by 0.0010 points in S_α and 0.0025 points in $F_\beta^{\max}$. This indicates that the backbone of the structure encoder plays a crucial role in our proposed dual-branch architecture.

Compared to other models (as shown in Table I), when considering both the ORSSD and EORSSD datasets comprehensively, the combination of Res2Net50 as the Image Encoder and ResNet18 as the Structure Encoder yields the most competitive results in both datasets. We would like to emphasize that our proposed model demonstrates superior results even with a smaller number of training images. In addition, by using a larger backbone, there is potential to achieve even higher performance on the EORSSD dataset, showcasing the model's scalability and adaptability to larger datasets.

6) Analysis on Failure Cases: We demonstrate excellent results in ORSI-SOD, but there are still challenges that need to be addressed. As shown in Fig. 11, we illustrate failure cases for each scenario. In the first row, our model fails to completely detect one of the multiple salient objects. This is due to the characteristics of the terrain. Although the objects belong to the same terrain category, their size and shape differ significantly, making it challenging for the model to detect them accurately. In the second row, our model shows inaccurate prediction results. Both images feature prominently cast shadows, which our model erroneously identifies as part of the salient object. The challenge arises because shadows form

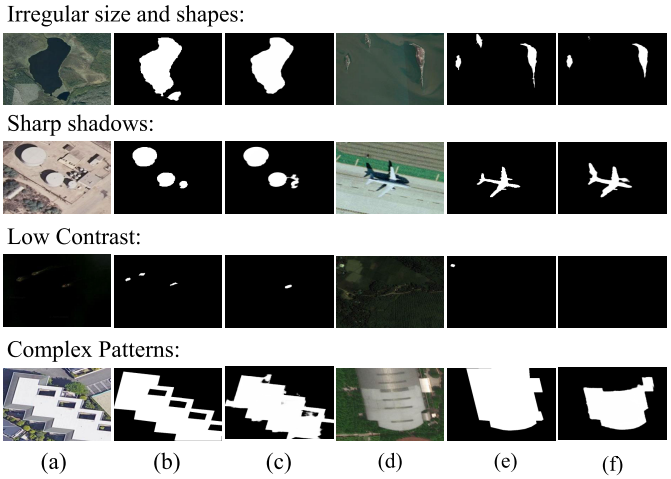


Fig. 11. Failure cases on our model. (a) ORSIs. (b) GTs. (c) Predicted results by Ours. Upper sentences of figures mean the case of challenging scenarios. (d) ORSIs. (e) GTs. (f) Predicted results by Ours.

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similar to the shape of the target object, making it difficult for the model to distinguish them explicitly. The third row contains images with low contrast. The low contrast blurs the boundaries between the object and the background, making it difficult for the model to predict accurately, especially when the target object is small. Our model completely failed to detect the target object due to the low contrast blurring the boundaries between the object and the background, and the small size of the target object. In the final row, the target building has a highly complex structure, making it challenging for our model to accurately predict its shape. Even when the target is sufficiently large, the complexity of its shape or variables such as color differences within the same building can hinder the model's ability to accurately detect it. Our model roughly detected the location, but it failed to capture the precise patterns.

As mentioned above, ORSI-SOD presents numerous challenges due to the wide variability in target shapes, sizes, forms, numbers, illumination, shadows, and resolution of ORSI. To develop a more precise model, extensive future research is required.

V. Conclusion

In this article, we address challenging aspects of ORSI-SOD, including irregular shapes and distributions, varying object sizes, and complex scenario issues. We propose a novel network, LSHNet. To leverage edge information in complex scenes, we introduce an edge encoder that deeply extracts edge information as a structure-prior. We propose ISFM to fuse the structure feature extracted from the structure-prior with the context features extracted from the image. To refine the feature map focused on the target object, we introduce LGFM, which extracts SCs representing the target object and integrates them with the features from ISFM. To maximize the efficiency of LGFM, we propose SCUM, which updates SCs of multilayers from high to low levels. We demonstrate the superiority of our proposed model on three datasets: ORSSD, EORSSD, and ORSI-4199.

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